

IAS 2026 Vancouver final scientific closure

Executive audit report | 2026-09-19 | not a submission paper or poster

Executive decision

Date: 2026-09-19

Executive decision

The final closure campaign does not clear the full award-level claim. It does produce a defensible, reproducible audit package with explicit negative and stopped results.

The strongest supported result is narrow: the frozen GFL11 device/port implementation is numerically consistent across Python and Julia over 64 canonical cases and 602 transfer points, and its network harness satisfies the declared residual limit. The official PowerDynamics and SimpleGFLDC IEEE-39 campaigns are valid alternative-model negative results. The official-model TDS cases solve and provide decay diagnostics without a reliable modal-frequency crossing.

The decisive limitation is that the exact Julia port of the frozen custom SynchronousMachine/first-order AVR/two-state IEEEEST PSS/no-governor model was not completed. The true same-model cross-code gate is therefore STOPPED_BY_GATE. Mechanism, physical robustness, genuine mixed holdout, and new scaling claims are not promoted downstream.

What survived

G0 frozen-artifact reproduction: FROZEN_ARTIFACT_REPRODUCED.

G1 canonical GFL11 Python/Julia device and transfer parity:
NUMERICALLY_VERIFIED.

G6 official PowerDynamics TDS execution: NUMERICALLY_VERIFIED, with scope limited to the official alternative model and decay diagnostics.

G7 official SimpleGFLDC second-model census: SECOND_MODEL_VALIDATED, with a 16/16 stable negative result.

G8 alternative-model holdout: BLIND_HOLDOUT, but single-class and not discriminative.

The claim ledger and reviewer attack matrix provide safe wording and adversarial scope controls.

What failed or remains unknown

Exact same-model SG/AVR/PSS Python-to-Julia parity: STOPPED_BY_GATE.

Full frozen IEEE-39 same-model base/V4 census: STOPPED_BY_GATE.

Collective mechanism test: NOT_TESTED because its prerequisite stopped.

Common physical uncertainty envelope, robust radius, and robust census:

NOT_TESTED.

New independent mixed holdout: not established; the executed holdout was single-class and needs a second preregistered holdout.

New asymptotic scaling sweep: NOT_TESTED.

Claim-safe conclusion

The package supports a scoped cross-code GFL11 parity result and documents model-conditioned negative results. It does not support the stronger statement that the frozen custom IEEE-39 collective blocker has been independently validated, refuted, robustly generalized, or predictively detected.

Artifact rule

This root contains no final paper or poster rewrite. The final ZIP contains only this executive report as a PDF plus raw evidence, code, reports, docs, figures, environment files, and hashes. Legacy generated paper/poster PDFs are explicitly excluded.

Gate adjudication

Gate	Status	Question	Observed
G0	FROZEN_ARTIFACT_REPRODUCED	frozen baseline and scope	nominal P4 reproduced; global 402/512 and targeted 511/512 retained
G1	NUMERICALLY_VERIFIED	canonical GFL11 Python/Julia parity	64 cases; 602 transfer points; max state relative error 5.14e-16
G2	STOPPED_BY_GATE	SG/AVR/PSS component parity	exact custom component port not completed
G3	STOPPED_BY_GATE	true frozen IEEE-39 base cross-code parity	not established
G4	STOPPED_BY_GATE	true frozen IEEE-39 V4 census	not established
G5	NOT_TESTED	collective mechanism	dependent on stopped same-model gate
G6	NUMERICALLY_VERIFIED	common-disturbance TDS	official-model base/one/full solves; no reliable modal crossing
G7	SECOND_MODEL_VALIDATE	SimpleGFLDC second-model census	16/16 initialized and stable
G8	BLIND_HOLDOUT	genuine alternative-model holdout	4/4 stable; single class
G9	NOT_TESTED	three-model comparison	not run after same-model gate
G10	NOT_TESTED	common physical uncertainty envelope	not computed
G11	NOT_TESTED	robust V4 census	not run
G12	NOT_TESTED	g-by-epsilon atlas	not run
G13	RETROSPECTIVE	topology calibration	12195 rows; 16 witnesses retained
G14	RETROSPECTIVE	governed P4 blocker	alpha 0.1270065 to -0.074541 retained
G15	RETROSPECTIVE	legacy transfer/scaling check	global and targeted V9 counts retained
G16	NOT_TESTED	new asymptotic scaling	no new sweep

The final ZIP deliberately excludes FINAL_PAPER.pdf and FINAL_POSTER_DRAFT.pdf. The true same-model gate is stopped, so downstream mechanism, robustness, mixed-holdout, and scaling claims are not promoted.